{3} Current Electricity

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* ELECTRIC CURRENT

* ELECTRIC CURRENTS IN CONDUCTORS

* OHM' S LAW

* DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY

* LIMITATIONS OF OHM 'S LAW

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* In the present chapter, we shall study some of the basic laws concerning steady electric currents.

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* The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm^2-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^{+} ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^{6} m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2\ A + (0.6\ A/s^2)\ t^2$.

(a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10$ s?

(b) What constant current would transpot the same charge in same charge in the same time interval?